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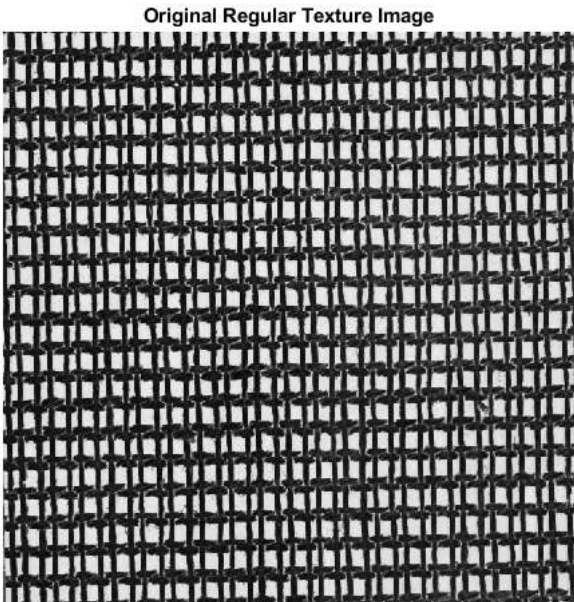
Experiment 10 : Find statistical moments and relative smoothness of texrure images.

```
clc; close all; clear;
```

Regular Texture

```
I_reg = double(imread('Texture/regular.gif'));  
mean_reg = mean2(I_reg);  
std_reg = std2(I_reg);  
var_reg = std_reg ^ 2;  
skew_reg = skewness(I_reg, 1, 'all');  
kurt_reg = kurtosis(I_reg, 1, 'all');  
rel_smooth_reg = 1 - (1/var_reg);  
  
figure,  
imshow(I_reg, []), title("Original Regular Texture Image");  
fprintf("Regular Texture : \nMean = %f, Variance = %f, Skewness = %f, Kurtosis = %f\n and Relative Smoothness = %f", mean_reg, var_reg, skew_reg, kurt_reg, rel_smooth_reg);
```

Regular Texture :
Mean = 103.857400, Variance = 7400.440377, Skewness = 0.483024, Kurtosis = 1.443703
and Relative Smoothness = 0.999865



Irregular Texture

```
I_irreg = double(imread('Texture/irregular.gif'));  
mean_irreg = mean2(I_irreg);  
std_irreg = std2(I_irreg);  
var_irreg = std_irreg ^ 2;  
skew_irreg = skewness(I_irreg, 1, 'all');  
kurt_irreg = kurtosis(I_irreg, 1, 'all');
```

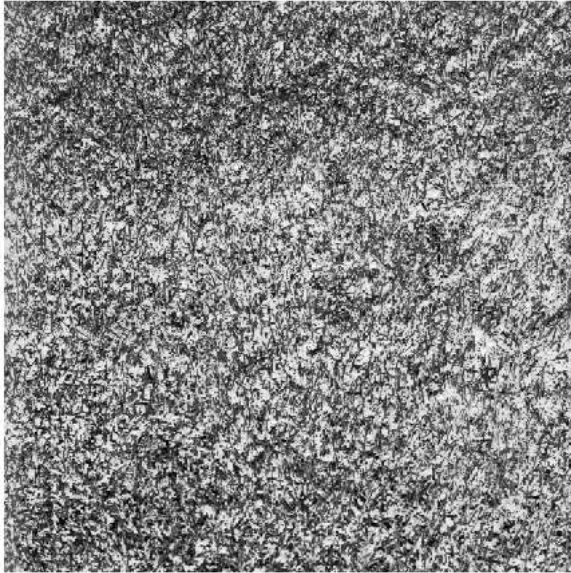
```
rel_smooth_irreg = 1 - (1/var_irreg);
```

```
figure,  
imshow(I_irreg, []), title("Original Irregular Texture Image");  
fprintf("\n\nIrregular Texture : \nMean = %f, Variance = %f, Skewness = %f, Kurtosis = %f and Relative Smoothness = %f", mean_irreg, var_irreg, skew_irreg, kurt_irr
```

Irregular Texture :

Mean = 130.991951, Variance = 3747.370004, Skewness = 0.074297, Kurtosis = 1.707152 and Relative Smoothness = 0.999733

Original Irregular Texture Image



Randomized Texture

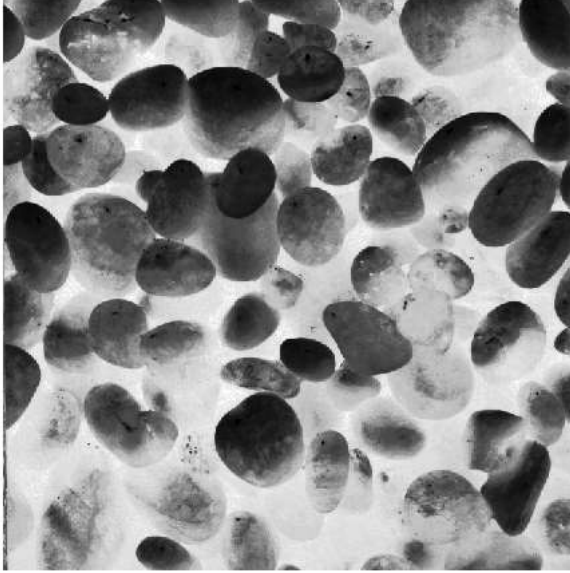
```
I_fin = double(imread('Texture/fin.gif'));  
mean_fin = mean2(I_fin);  
std_fin = std2(I_fin) * 255;  
var_fin = std_fin ^ 2;  
skew_fin = skewness(I_fin, 1, 'all');  
kurt_fin = kurtosis(I_fin, 1, 'all');  
rel_smooth_fin = 1 - (1/var_fin);
```

```
figure,  
imshow(I_fin, []), title("Original Randomized Texture Image");  
fprintf("\n\nRandomized Texture : \nMean = %f2, Variance = %f, Skewness = %f, Kurtosis = %f and Relative Smoothness = %f\n", mean_fin, var_fin, skew_fin, kurt_fin,
```

Randomized Texture :

Mean = 137.1229172, Variance = 319178016.864534, Skewness = -0.163418, Kurtosis = 1.560401 and Relative Smoothness = 1.000000

Original Randomized Texture Image



Conclusion

First 4 statistical moments and Relative Smoothness were determined for different types of textures. It is observed that all texture images have high variance and relative smoothness values (higher than 0.93) indicating high variations in intensities.